OPERATING SYSTEMS – ASSIGNMENT 1

TASK 8 – Evaluate CPU States and Protection (3 Marks)
CO1–CO5 | Bloom's Level: K5 – Evaluate

Introduction

The operating system and CPU hardware work together to protect programs and control access to important resources. User mode and kernel mode prevent ordinary applications from directly performing privileged operations.

1. User mode and kernel mode

User mode: Applications run with limited privileges. They cannot directly execute privileged instructions or access protected kernel memory.
Kernel mode: The operating-system kernel runs with higher privileges. It can manage memory, devices, interrupts and processes.
Mode bit: A hardware status bit indicates the current execution mode. A system call or interrupt can transfer execution to kernel mode.

2. Privileged instructions

Privileged instructions are operations that must only be executed by the kernel, such as controlling interrupts or changing memory-management settings. If a user program attempts an unauthorised privileged instruction, the hardware raises an exception and the OS handles it.

3. Interrupts, traps and timer interrupts

An **interrupt** is a signal that requests CPU attention, often from a hardware device. A **trap** is a synchronous event caused by the currently executing instruction, such as a system call or an error. A **timer interrupt** occurs at a programmed time and allows the OS to regain control of the CPU, update scheduling information and switch tasks when needed.

4. What if any program could disable interrupts?

If ordinary programs could disable interrupts, a faulty or malicious program might prevent the OS from responding to timer events and hardware signals. The CPU could remain occupied by that program, and patient-monitor tasks might not run on time. Therefore, disabling interrupts must be a privileged operation controlled by the kernel.

5. How hardware protects memory

Memory-management hardware, such as the memory management unit (MMU), checks each memory access against the process's permitted address space and access rights. Page tables define which memory pages a process can access. An invalid access causes a fault, which the OS handles. This prevents one ordinary program from freely reading or modifying another program's memory.

6. Alternative concept: polling instead of interrupts

With polling, the monitor repeatedly checks the sensor at fixed intervals. If the polling interval is T, an abnormal reading that occurs just after a check may wait almost T before it is noticed, plus processing time. A shorter interval reduces the maximum waiting time but causes more CPU activity and power use. Interrupt-driven I/O allows the sensor or device to notify the CPU when an event occurs, so the CPU does not need to check continuously. Interrupt latency is not zero and must also be measured and bounded for a real-time system. For the hospital's 10 ms alarm requirement, an appropriately designed interrupt-driven real-time approach is preferable to slow polling.

Conclusion

Hardware privilege levels, interrupts and memory protection are essential for a safe hospital system. Patient monitors need predictable scheduling and protected access to ensure timely alarms.

Prepared from the corresponding task in the uploaded CSA04 Operating Systems Assignment 1 document. Add your name, register number, required screenshots and your own measured evidence wherever requested before submission.